2022-23
Mechanics
(Circular motion II)

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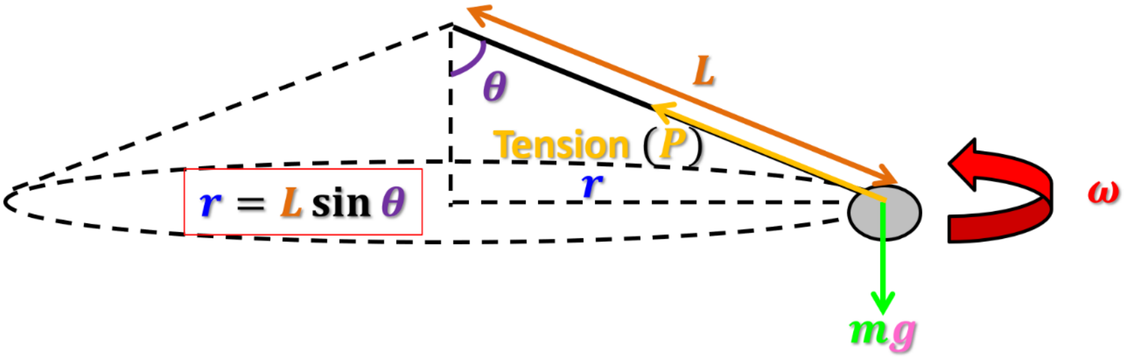
(i) Conical pendulum

We use the letter **P** for the **tension** to avoid confusion with the period **T**.



Consider a conical pendulum with an object of mass m, circulating with radius at an angular speed ω.

Centripetal force is provided by horizontal component of tension P.

Horizontal component:

$$P\sin\theta = mr\omega^2$$

Vertical component:

$$P\cos\theta = mg$$

An expression for angular velocity ω of the conical pendulum:

$$\because \quad \tan\theta = \frac{r\omega^2}{g}$$

$$\therefore \quad \frac{\sin\theta}{\cos\theta} = \frac{L\sin\theta\,\omega^2}{g}$$

$$\therefore \quad \omega = \sqrt{\frac{g}{L\cos\theta}}$$

Period T of circular motion of conical pendulum:

$$T = 2\pi\sqrt{\frac{L\cos\theta}{g}}$$

1. A bob of mass 0.1 kg is suspended by a string of length 80 cm from a ceiling. The bob moves in a horizontal circle and the sting makes an angle of 20° with the vertical. (Take $g = 10\ \text{ms}^{-2}$)
   (a) Find the tension in the sting
   (b) Find the angular speed of the bob
   (c) Find the period of the circular motio

(a) $T\cos\theta = mg$

$T\cos 20° = 0.10(10)$

$T = 1.06N$

(b) $F = mr\omega^2$

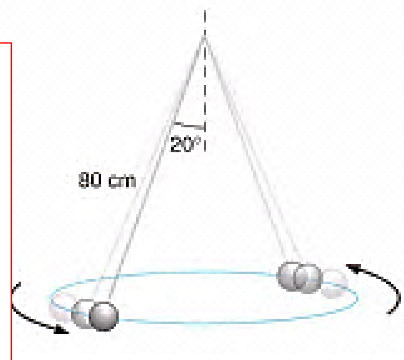
$1.06\sin 20° = (0.1)(0.8\sin 20°)\omega^2$

$\omega = 3.65rads^{-1}$

(c) $\omega = \frac{2\pi}{T}$

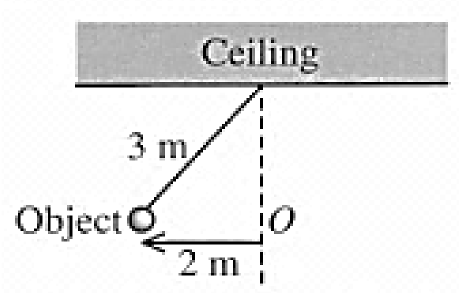
$3.65 = \frac{2\pi}{T}$

$T = 1.72s$

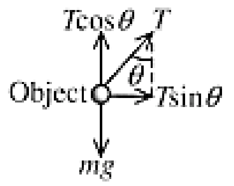


2. An object of mass 0.5 kg was tied to the ceiling by a 3 m long inextensible string. The object rotated along a horizontal circular path of radius 2 m. The speed of the object was v. Point O was the centre of the circular path. (Take $g = 10\ \text{ms}^{-2}$)

   (a) Draw the free body diagram of the object.
   (b) Find the tension of the string.
   (c) Find the centripetal force experienced by the object.
   (d) Find the value of v.
   (e) After some time, the object fell and rotated along a lower horizontal circular path. Describe the change of the tension.



(a)



(b) $\sin\theta = \dfrac{2}{3}$, $\theta = 41.8^\circ$

$T\cos\theta = mg$

$T = \dfrac{0.5\times10}{\cos 41.8^\circ} = 6.71$ N

(c) $F_C = T\sin\theta$

$F_C = 6.71\sin 41.8^\circ = 4.47$ N

(d) $F_C = \dfrac{mv^2}{r}$

$v = \sqrt{\dfrac{rF_C}{m}} = \sqrt{\dfrac{2\times4.47}{0.5}}$

$v = 4.23\ \text{m s}^{-1}$

(e) As θ decreased, the vertical component of the tension increased. But the magnitude of this component remained equal to mg. Thus the tension decreased.

3. A pendulum bob of mass m tied to a string of length l is set to swing in a horizontal circle. The string makes an angle of 45° with the vertical. What are the (a) tension in the string and (b) the speed of the bob?

   A.
   B.
   C.
   D.

(a) Let T be the tension in the string.
Since vertical forces are balanced,

$T\cos 45^\circ = mg$

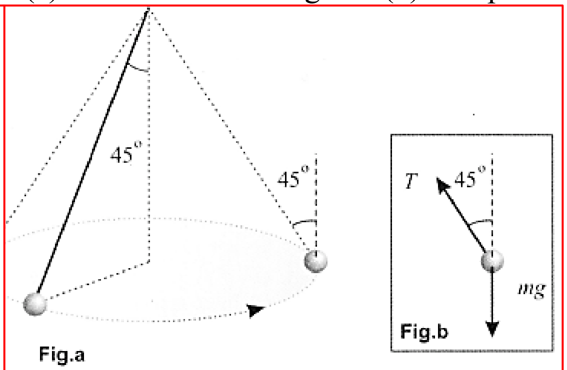
$\therefore\ T = \sqrt{2}mg$

(b) The radius of the circle is $r = l\sin 45^\circ$.
The horizontal component of the tension provides the centripetal force. Thus, the speed of the bob is given by

$T\sin 45^\circ = \dfrac{mv^2}{r}$

$v^2 = \dfrac{rT\sin 45^\circ}{m} = \dfrac{\sqrt{2}mgl\sin^2 45^\circ}{m} = 0.707gl$

$\therefore\ v = 0.84\sqrt{gl}$



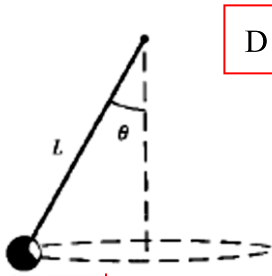
D

4. The figure shows a small body of mass m revolving in a horizontal circle with a constant speed v at the end of a string of length L. If the string makes an angle θ with the vertical, what is the period of the body?

   A. $\dfrac{\pi}{2}\sqrt{\dfrac{L\sin\theta}{g}}$ B. $\pi\sqrt{\dfrac{L}{g\cos\theta}}$

   C. $2\pi\sqrt{\dfrac{L\sin\theta}{g}}$ D. $2\pi\sqrt{\dfrac{L\cos\theta}{g}}$

D



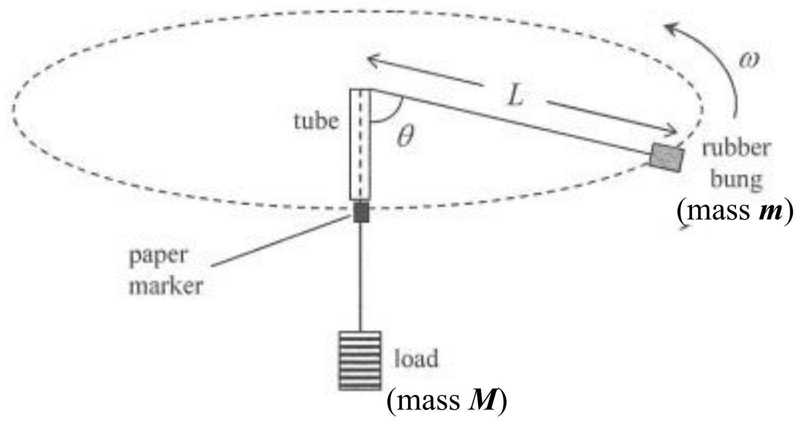
The figure shows a small body...

D In the vertical direction, $T\cos\theta = mg$. In the horizontal direction, $T\sin\theta = mr\omega^2$, where $r = L\sin\theta$. Solving, we get $\omega^2 = g/(L\cos\theta)$.

$\therefore$ period $= \dfrac{2\pi}{\omega} = 2\pi\sqrt{\dfrac{L\cos\theta}{g}}$

Alternative: use $T\sin\theta = mv^2/r$, instead.

Experimental test of centripetal force



Consider the load with mass M:

$$P = Mg$$

Consider the rubber bung with mass m:

$$P\cos\theta = mg$$

$$P\sin\theta = mL\sin\theta\,\omega^2$$

Consider the angle of the string made with the vertical:

$$\because \quad Mg\cos\theta = mg$$

$$\therefore \quad \cos\theta = \frac{m}{M}$$

(the angle depends on m and M but independent of the speed v, angular speed ω and the length L)

Sources of error:

- There is friction between the string and the tube.
- It is difficult to maintain a constant speed of the rubber bung.
- The rubber bung may not be swirled in a horizontal circular motion.
- The load may not be at rest during the whirling of the rubber bung.

- The tube is held vertically and the rubber bung is whirled into a horizontal circle. The speed of the bung is increased until the marker is just below the table.
- Since the tension is equal to the weight of the load (screw nuts), the tension is constant.
- As the vertical component of the tension balances the weight of the rubber bung, the angle θ is also constant.
- The horizontal component of the tension provides the centripetal force.

Remarks: It takes less than 1 s for rubber bung to complete 1 revolution. Hence, time taken for many revolutions (e.g.: 50), instead of 1 revolution is measured to reduce the percentage error in measurement.

5. A student performs a centripetal force experiment using a rubber bung tied to one end of a string which passes through a glass tube with smooth openings and has a weight W hanging at its other end. The rubber bung is set into a horizontal uniform circular motion with angular speed ω while the length of the string beyond the upper opening of the glass tube is L. The portion of the string attaching to the rubber bung makes an angle θ with vertical.

C

Which of the following statements is/are true?

(1) θ is independent of ω.

(2) If a larger weight W is used, the angle of Inclination θ is larger for the

(3) L is independent of ω.

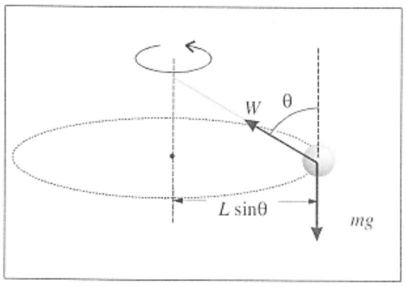
A. (1) only

B. (3) only

C. (1) and (2) only

D. (2) and (3) only

The tension in the string is exactly equal to W because the screw nuts are in equilibrium. The force diagram for the rubber bung is shown below:



(1) is correct. The vertical level of the rubber bung stays the same. It does not have vertical motion. The vertical forces of the bung are balanced. Therefore, the angle of the string is given by

$$W\cos\theta = mg$$

$$\therefore \quad \cos\theta = \frac{mg}{W} \quad \text{..........(1)}$$

Clearly, θ is independent of angular speed ω.

(2) is correct. From equation (1), as W increases, cos θ decreases. This means that θ increases.

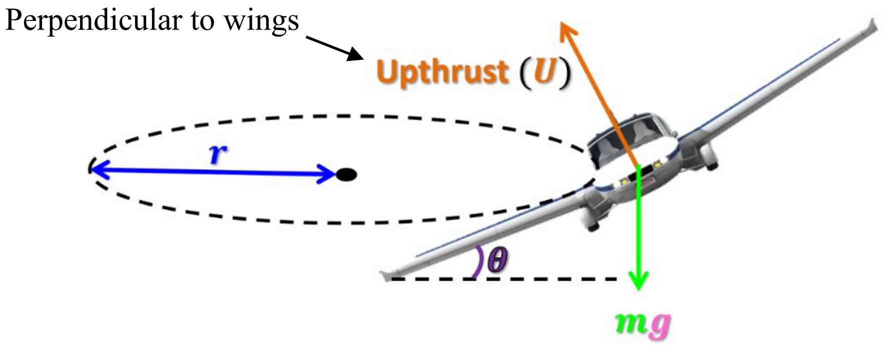
(3) is not correct. Consider the horizontal circular motion. The horizontal component of W provides the centripetal force:

$$W\sin\theta = mr\omega^2 = m \times L\sin\theta \times \omega^2$$

$$\therefore \quad L = \frac{W}{m\omega^2} \quad \text{........(2)}$$

L is smaller for higher ω.

(ii) Aircraft making a turn



Horizontal component:

$$U\sin\theta = \frac{mv^2}{r}$$

Vertical component:

$$U\cos\theta = mg$$

Tilting angle:

$$\tan\theta = \frac{v^2}{gr}$$

When an aeroplane flies in the air, the lifting force (upthrust) acting on it is normal to the wings. In order to turn, the aeroplane has to incline to the horizontal.

Remarks:

- Centripetal force is provided by **horizontal component of upthrust**.
- Speed v is independent of mass m.

6. An aircraft flies along a horizontal circular path of radius 20 km with a constant speed of 250 ms^{-1}.
   (a) Find the angle that the upthrust makes with the vertical. (Take $g = 10\ \text{ms}^{-2}$)
   (b) If the mass of the aircraft is 2000 kg, what is the upthrust?

a. 17.4°
b. 21000 N

7. A toy aeroplane of mass 0.5 kg flies in a horizontal circle. The lifting force $U = 10$ N is perpendicular to the wings. The aeroplane completes one revolution in 6 s.
   (a) Find the angular speed of the aeroplane. (2 marks)
   (b) Find θ, the angle between the vertical and the lifting force (2 marks)
   (c) Find the radius of the circle r. (2 marks)
   (d) The lifting force U remains unchanged. How does θ change if
      (i) the mass of the aeroplane becomes smaller.
      (ii) the aeroplane travels at a higher speed but the radius of the circle remains unchanged. (2 marks)

(a) The angular speed is $\frac{2\pi}{T} = \frac{2\pi}{6} = 1.047 \approx 1.05\ \text{rad s}^{-1}$. (1M+1A)

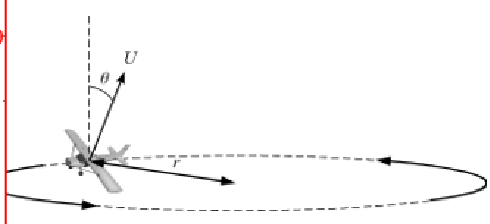
(b) The vertical component of the lifting force balances the weight of the aeroplane, i.e.

$$U\cos\theta = mg$$
$$10 \times \cos\theta = 0.5 \times 10$$
$$\theta = 60°$$ (1M+1A)

(c) The horizontal component of the lifting force provides the centripetal force, i.e.

$$U\sin\theta = mr\omega^2$$
$$10 \times \sin 60° = 0.5 \times r \times (1.047)^2$$
$$r \approx 15.8\ \text{m}$$ (1M+1A)

(d) (i) increases (1A)
    (ii) increases (1A)



8. An aeroplane is turning left at a constant speed of $175\ \text{ms}^{-1}$, it flies along a horizontal circular path with radius of $20\ \text{km}$. What is the angle between its wings and the horizontal?

A. 8.2°
B. 8.8°
C. 9.2°
D. 9.7°

B
Net force for circular motion

$= F\sin\theta = \dfrac{mv^2}{r}$ .... (1)

The weight is balanced by the vertical component of F,

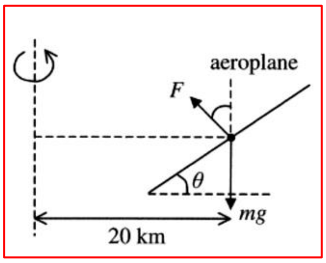
$F\cos\theta = mg$........ (2)

(1) ÷ (2)

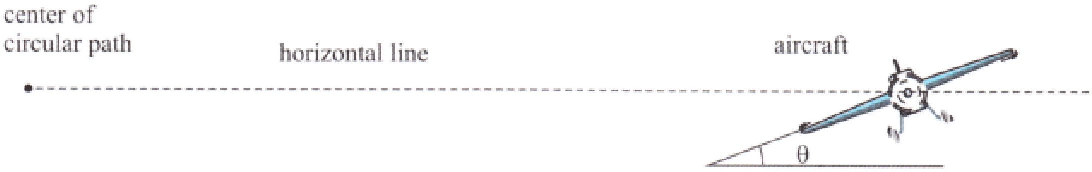
$\tan\theta = \dfrac{v^2}{gr}$

$= \dfrac{(175)^2}{(10)(20\times10^3)}$

$\therefore\ \theta = 8.7°$



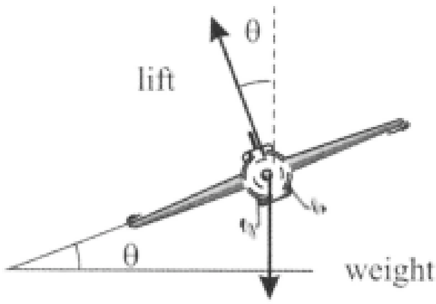
9. An aircraft is flying at a constant speed of $500\ \text{km h}^{-1}$ in a horizontal circle of radius $4\ \text{km}$ as shown.



(a) Explain why the aircraft has a constant speed but is accelerating.
(b) To turn around, the aircraft is banked as shown. State how this produces the centripetal force necessary for the circular motion.
(c) (i) Calculate the acceleration experienced by the pilot as the aircraft turns.
(ii) Calculate the angle θ through which the airplane must tilt in order to perform the circular motion.
(d) If the mass of the airplane is $2500\ \text{kg}$, what must be the lift produced by the wing?

(a) Although the speed of the aircraft is constant, its direction is changing continuously. As the velocity is changing, the aircraft is accelerating.

The lift of the aircraft acts normally to the wing. As the aircraft is banked, the horizontal component of the lift becomes the net force that accounts for the centripetal force.



(c) i) The centripetal acceleration is

$$a = \frac{v^2}{r} = \frac{\left(\frac{500\times1000}{3600}\right)^2}{4000}$$

$= 4.82\ \text{ms}^{-2}$

ii) Let L be the lift of the aircraft.
Equating forces vertically,

$L\cos\theta = mg$ ....(1)

As the horizontal component of L provides the centripetal force,

$L\sin\theta = \dfrac{mv^2}{r}$ ....(2)

$\tan\theta = \dfrac{v^2}{rg} = \dfrac{a}{g} = \dfrac{4.82}{10}$

$\therefore\ \ \theta = 25.7°$

(d) From equation (1) above, the lift is

$L = \dfrac{mg}{\cos\theta} = \dfrac{25000}{\cos 25.7°}$

$= 2.78\times10^4\ \text{N}$

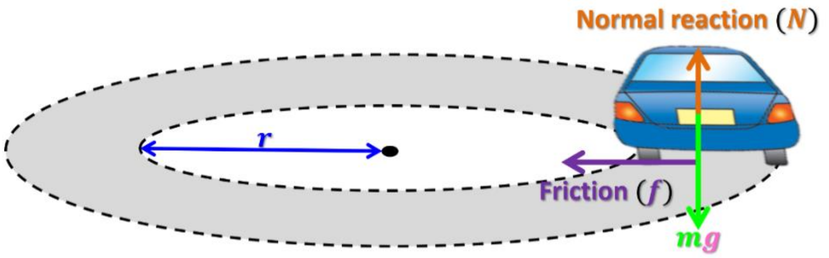
(iii) Rounding a bend

Horizontal component:

$$f = \frac{mv^2}{r}$$

Vertical component:

$$N = mg$$



Remarks:

- Centripetal force is provide by **friction**
- The greater the speed v or the smaller the radius r, the larger is the friction required. The friction between the road and the tyres of the car has a maximum value f_{max}. If the required centripetal force is larger than f_{max}, skidding will occur.

(iv) Turning round on ideal banked road

Horizontal component:

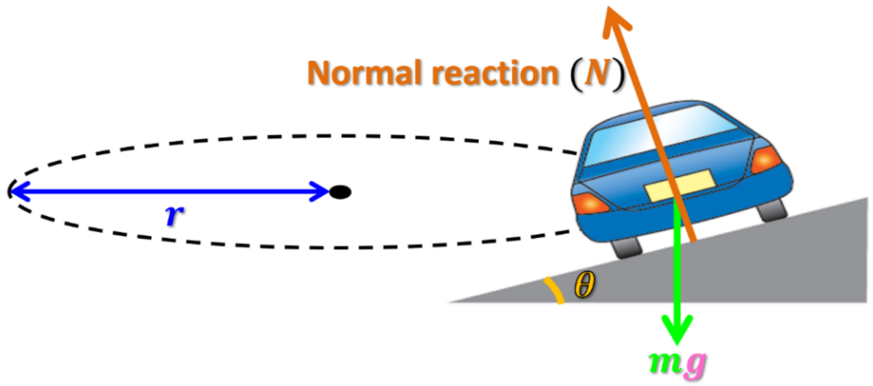
$$N \sin\theta = \frac{mv^2}{r}$$

Vertical component:

$$N \cos\theta = mg$$

Ideal banking angle:

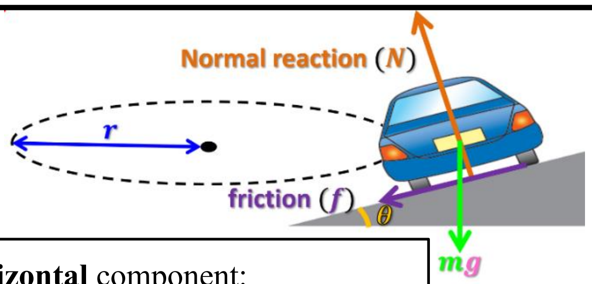
$$\tan\theta = \frac{v^2}{gr}$$



Remarks:

- Banking road helps the vehicle to prevent skidding during turning round a corner.
- Ideal banked road is designed so that no friction is required for circular motion. Centripetal force is provided by **horizontal component** of normal reaction.

(a) Turning round on banked road with high speed



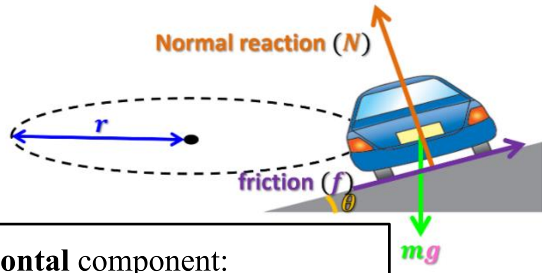
Horizontal component:

$$N \sin\theta + f\cos\theta = \frac{mv^2}{r}$$

Vertical component:

$$N \cos\theta = mg + f\sin\theta$$

(b) Turning round on banked road with low speed



Horizontal component:

$$N \sin\theta - f\cos\theta = \frac{mv^2}{r}$$

Vertical component:

$$N \cos\theta + f\sin\theta = mg$$

10. 2015 DSE MC Q.10

The figure shows the rear view of a car of mass m which travels along a circular road banked with an angle θ to the horizontal. The car moves at a certain speed such that it experiences **no frictional force along the inclined surface.** Which of the following represents the centripetal force on the car ?

A. $mg\sin\theta$

B. $mg\sin\theta\cos\theta$

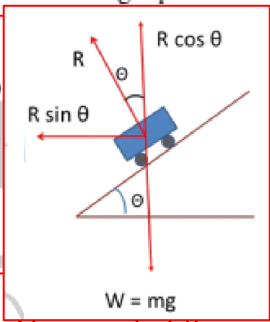
C. $\dfrac{mg\cos\theta}{\sin\theta}$

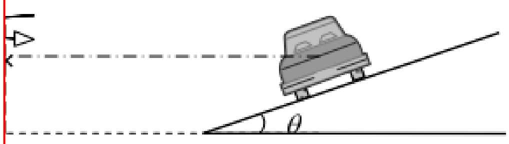
D. $\dfrac{mg\sin\theta}{\cos\theta}$

D

- **Vertical Direction:** $R\cos\theta = mg \cdots(1)$
- **Horizontal Direction:** $R\sin\theta = \dfrac{mv^2}{r} \cdots(2)$
- **By (1),** $R = \dfrac{mg}{\cos\theta} \cdots(3)$
- **Therefore, The centripetal force**

$= (\dfrac{mg}{\cos\theta})\sin\theta = \dfrac{mg\sin\theta}{\cos\theta}$





11. A car turns a corner on a banked road as shown. The speed of the car is lower than the average road speed If N is the normal force acting on the car by the road and f is the friction acting on the car along PQ, the centripetal force of the car is

A. $N\sin\theta + f\sin\theta$

B. $N\cos\theta + f\cos\theta$

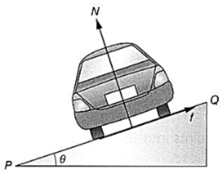
C. $N\cos\theta - f\cos\theta$

D. $N\sin\theta - f\cos\theta$

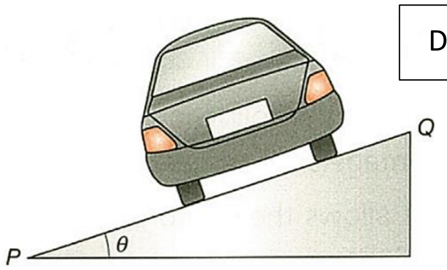
D

The forces acting on the car are mg, N and f. Since mg is a constant and N is perpendicular to $F_{//}$, f must point in the direction that makes $F_{//}$ decrease.

∴ f points towards Q.



∴ Centripetal force = $N\sin\theta - f\cos\theta$



12. A car of mass m travels with a constant speed round a curved level track with radius of curvature r, as shown in the figure below.

(a) Draw the direction of the following on the diagram:

(i) The net force F acting on the car (1 mark)

(ii) The velocity v of the car (1 mark)

(b) (i) Explain why the car is accelerating although its speed is constant. (1 mark)

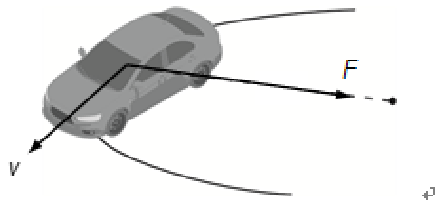
(ii) State the direction of its acceleration. (1 mark)

(c) What is the work done by F? Explain your answer. (2 marks)

(d) It is given that m = 1200 kg, v = 12.5 m s^{-1} and r = 25 m. Calculate the net force acting on the car. (2 marks)

(e) Sometimes, curved track are banked instead of level. Explain the advantage of banking. (2 marks)

(a)



(i) (Correct F)

(ii) (Correct v)

(b) (i) The direction of the car's velocity is changing.

(ii) Towards the centre of curvature of the track

(c) F is perpendicular to the direction of motion of the car. Therefore, it does zero work.

(d) Net force = centripetal force

$= \dfrac{mv^2}{r}$

$= \dfrac{1200 \times 12.5^2}{25}$

= 7500 N

(e) The normal force from the track helps provide the centripetal force. This considerably increases the limiting speed for making a safe turn.

13. In Figure 1, a car of weight W is moving with uniform speed on a banked road along a circular path of horizontal radius 10 m. The angle of inclination of the road is 20°. Assume that the centripetal force arises entirely from a component of the normal reaction N from the road.

(a) Draw and label all the forces acting on the car in Figure 1 at the instant shown. (2 marks)

(b) Given that the weight of the car W is 20000 N

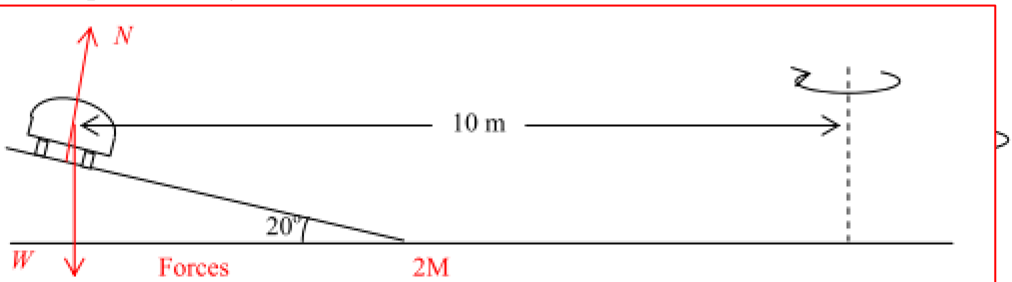
(i) By considering the vertical force on the car, find the value of N from the given data. (2 marks)

(ii) Hence, find the speed of the car (2 marks)

(c) It is known that the car can be kept at the same height of the track even if the speed of the car is suddenly reduced along its track. Explain briefly (2 marks)

(a)

Figure 1



Forces 2M

(b)(i)

$20\,000 = N\cos 20^{\circ}$ 1M

$N = 21283.6$ N 1M

(b)(ii)

$(21283.6)\sin 20^{\circ} = (20\,000/9.81)v^2/(10)$ 1M

$v = 5.98$ m s^{-1} 1M

(c)

Centripetal force required is reduced as v is smaller, 1M

Friction acts upslope to reduce centripetal force. 1M

14. A car of mass 1200 kg is travelling at constant speed along a road which is an arc of a circle of radius 80 m. The road is banked at 15° to the horizontal.

At on particular speed v of the car, there is no frictional force acting on the tyres at 90° to the direction of motion of the car.

(a) On Figure (a), show all the forces acting on the car.

(b) Find the magnitude of the resultant force acting on the car.

(c) What should be the value of v?

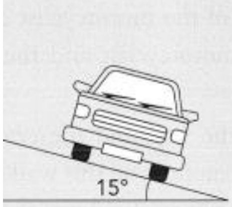


Figure (a): front view

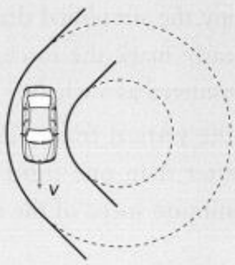
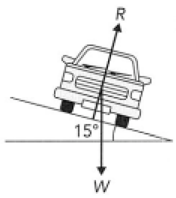


Figure (b): top view

(a) R = normal reaction
W = weight of the car



(b) Forces in the vertical direction are balanced.

$R\cos 15^{\circ} = W$

$= (1\,200)(9.81)$

$R = \dfrac{11\,772}{\cos 15^{\circ}}$ N

Resultant force

$= R\sin 15^{\circ}$

$= \dfrac{11\,772}{\cos 15^{\circ}} \times \sin 15^{\circ}$ *(1)*

$= 3.15 \times 10^{3}$ N *(1)*

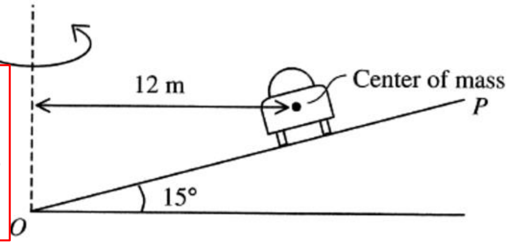
(c) $m\dfrac{v^2}{r} = R\sin 15^{\circ}$

$1\,200 \times \dfrac{v^2}{80} = \dfrac{11\,772}{\cos 15^{\circ}} \times \sin 15^{\circ}$ *(1)*

$v = 14.5$ m s^{-1} *(1)*

15. A car moves along a circular path with a radius of 12 m on a banked road of inclination 15° as shown in the figure. If there is no friction acting on the car along *OP*, what is the speed of the car?

A. $5.2\ \text{ms}^{-1}$
B. $5.4\ \text{ms}^{-1}$
C. $5.5\ \text{ms}^{-1}$
D. $5.6\ \text{ms}^{-1}$



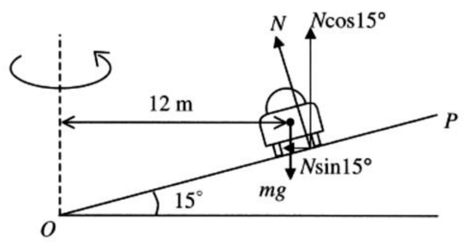
D

For no friction acting on the car along *OP*.

$N\sin 15° = \dfrac{mv^2}{r}$ .... (1)

(only the horizontal component of normal reaction provides the centripetal force)

$N\cos 15° = mg$ .... (2) (no vertical acceleration)



(1) ÷ (2), $\tan 15° = \dfrac{v^2}{gr}$

$\tan 15° = \dfrac{v^2}{(10)(12)}$

$v = 5.7\ \text{ms}^{-1}$

16. A man is riding a bicycle around a corner on a horizontal level. The man has to lean on one side. Which of the following statement is/are correct?

(1) He has to lean towards the centre of curvature of the corner.
(2) He has to lean so that his weight can provide an extra centripetal force.
(3) He has to lean more if he rounds the corner at a higher speed.

A. (1) only B. (2) only
C. (1) and (3) only D. (2) and (3) only

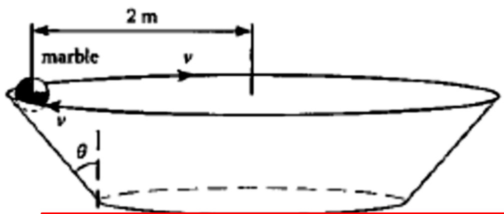
C

A man is riding a bycycle...
C (2) is incorrect because his weight has no horizontal component.

17. In the figure as shown, a marble of mass 0.01 kg moves around a horizontal circular track which makes an angle θ with the vertical. Assume that the marble is in uniform circular motion and the centripetal force is provided by the normal reaction from the track ONLY. Which of the following expressions is correct? Neglect friction.

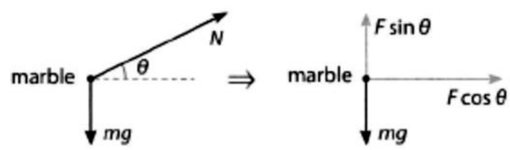
A. $\tan\theta = \dfrac{v^2}{2g}$ B. $\tan\theta = \dfrac{2g}{v^2}$

C. $\tan\theta = \dfrac{0.1\times 2g}{v^2}$ D. $\tan\theta = \dfrac{v^2}{0.1\times 2g}$



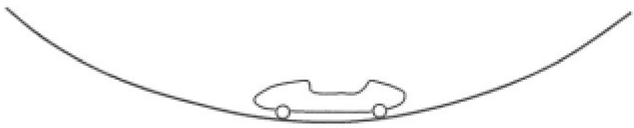
B

In the figure as shown, a marble...
B In the vertical direction, $N\sin\theta = mg$, where N is the normal reaction. In the horizontal direction, $N\cos\theta = mv^2/r$. This gives $\tan\theta = gr/v^2$.



■ The equations for the two components look alike, but they have completely different meanings:
Vertical: net force $F_y = 0$.
Horizontal: net force $F_x = mv^2/r$.

18. A car of mass m travels into a region where the track is an arc of a vertical circle of radius r. At the bottom of this arc, the car travels at speed v. At this position the vertical force exerted upwards by the track on the car is



A. $\dfrac{mv^2}{r}$

B. mg

C. $\dfrac{mv^2}{r} - mg$

D. $\dfrac{mv^2}{r} + mg$

D

The vertical force exerted on the car by the track is the normal reaction R.

The centripetal force is the net force towards the centre, thus the centripetal force is $R - mg$.

$\therefore\ R - mg = \dfrac{mv^2}{r}$ $\therefore\ R = mg + \dfrac{mv^2}{r}$

(v) Rotor in amusement park

In the horizontal direction:

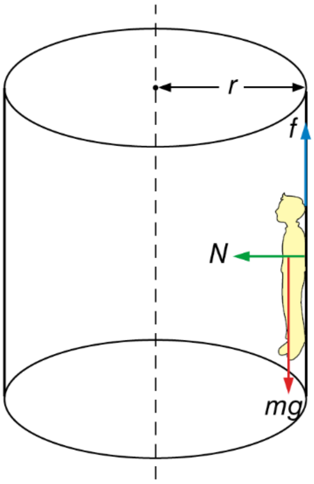
The normal reaction N acting on a rider by the wall provides the centripetal force

$$N = \frac{mv^2}{r}$$

In the vertical direction:

The weight of the rider must be balanced by the friction between the rider and the wall

$$f = mg$$



Remarks: f_{max} increase with normal reaction, which in turn increase with rotating speed. There is a minimum rotating speed to keep a rider stuck to the wall.

19. When a rotor of radius 2 m rotates, a man (60 kg) sticks to the wall without falling. The maximum friction f_{max} between the man and the wall depends on the normal reaction N acting on the man by the wall and is given by $f_{max} = 0.5N$.

(a) Find the friction between the man and the wall of the rotor. (1 mark)

(b) Suppose the rotor rotates at an angular speed such that the man just does not fall from the wall.

(i) what is the angular speed of the rotor (2 marks)

(ii) what is the period of the rotor (1 mark)

(iii)(1) What would happen to the man if the angular speed of the rotor increase?

(2) What would happen to the man if the angular speed of the rotor decrease?

(3) Can a man of larger mass remain stuck to the wall in a rotating rotor? Explain your answer.

(a)

Friction $f = mg$

$= 60 \times 9.81$

$= 588.6 \text{ N} \approx 589 \text{ N}$

(b)(i)

Centripetal force $F = N$

$= \frac{f}{0.5}$

$= \frac{588.6}{0.5}$

$= 1177 \text{ N} \approx 1180 \text{ N}$

By $F = mr\omega^2$,

$\omega = \sqrt{\frac{F}{mr}} = \sqrt{\frac{1177}{60 \times 2}} = 3.132 \text{ rad s}^{-1} \approx 3.13 \text{ rad s}^{-1}$

(b)(ii)

Period $= \frac{2\pi}{\omega} = \frac{2\pi}{3.132} = 2.01 \text{ s}$

(iii)(1)

The man would still stick to the wall.

(iii)(2)

$N\downarrow$ and f_{max} would become smaller than mg.

$\Rightarrow$ The man would fall.

(iii)(3)

Consider a rider of mass m who just does not fall from the wall.

f_{max} acting on him $= 0.5N = 0.5mr\omega^2$

f_{max} balances the his weight.

$f_{max} = mg$

$0.5mr\omega^2 = mg$

$\omega = \sqrt{\frac{g}{0.5r}}$

$\therefore$ The angular speed that keeps him stuck to the wall is not affected by his mass.

$\therefore$ The man of a larger mass can remain stuck to the wall.